ELECTRICAL ENGINEERING & SMART GRID INFRASTRUCTURE

Square-Mile AI Data Centres Requiring GW-Scale Power Connections

From hyperscale to "AI Factory" — engineering the electrical backbone, managing load volatility, bridging the grid infrastructure gap, and building resilient campuses for India's AI future.

GW-Scale Grid Interconnect | Captive Renewables & BESS | India AI Infrastructure | Load Variability & Grid Stability | Future: Space Data Centres

Senior Data Centre Electrical Design Engineer | Smart Grid & HV Infrastructure Specialist

Panel Session — March 2026

The Scale Shift: AI Factories & India's GW Ambition

📈 POWER DEMAND PROGRESSION

Classic Colo	5 MW
Hyperscale DC	50 MW
Cloud Campus	200 MW
AI Campus	500 MW
AI Factory	1–2 GW

Footprint: up to **640+ acres (1 sq. mile)**. Requires dedicated 400 kV transmission corridors — not shared distribution feeds.

💣 INFRASTRUCTURE TIMELINE GAP — CRITICAL RISK

DC Construction	6–18 mo
New Generation	9–24 mo
Transmission	3–6 yrs

⚠️ Data centres commission faster than grid reinforcements can follow — grid planning must precede land acquisition.

IN INDIA GROWTH TRAJECTORY

- **2020:** 520 MW — colocation-focused, Mumbai/Chennai dominated
- **2025: 1.5 GW** (3× growth). 1.02B internet subscribers. 25 GB/user/month. $95B invested 2019–2025
- **Key deals:** AdaniconneX + Google $15B (Vizag). Reliance 1 GW compute campus.
- **2030:** >10 GW target. Shift to coastal greenfield with co-located renewables & submarine cables.

10 GW India target 2030	**$95B** Investment 2019–25
400 ZB Global data by 2028	**2.7×** Data growth vs 2024

🏢 TOP 5 OPERATORS (66%+ SHARE)

ST Telemedia · NTT · Sify · CtrlS · Nxtra (Airtel). Growth moving from Tier-1 metros toward purpose-built AI Park SEZs with pre-approved grid infrastructure and fast-track environmental clearances.

Load Variability & Grid Stability: The Central Engineering Challenge

THE LOAD SWING PROBLEM

GPU training jobs start, checkpoint, and stop — creating severe swings the national grid was never designed to absorb:

Training burst swing **500 MW → 100 MW**

±400 MW swing per campus — seconds to minutes

- Frequency nadir: RoCoF exceeds CERC norms (>0.5 Hz/s)
- Voltage regulation collapse on shared 220/400 kV buses
- Synchronisation threat to co-located rotating generators

MANDATORY CLUSTER STUDY REQUIREMENTS

DP 2

Before any campus >200 MW connects, CERC/SERC must mandate:

- **Load Flow (N-1/N-2):** Steady-state contingency across 400 kV network
- **Dynamic Stability:** Transient & voltage stability with realistic GPU load profiles — not flat demand
- **SCL Assessment:** Short Circuit Level from on-site generation interaction with grid fault impedance
- **Harmonic Study:** THD impact from massed power electronics — IEEE 519 / IEC 61000-3
- **RoCoF Assessment:** Worst-case instant loss of full campus load on shared bus

LOAD SENSITIVITY & LVRT FAILURE

GPU PSUs and UPS systems lack adequate **Low Voltage Ride-Through (LVRT)** capability. During grid disturbances: mass DC load trips → sudden loss of hundreds of MW → frequency overshoot. Cascading UPS recharge surge creates a second grid impact after fault clearance. Harmonic injection from rectifier banks degrades power quality for co-located loads.

Regulatory Gap (DP 4): India's CERC regulations and NLDC/RLDC operating procedures were not designed for single loads >200–300 MW. Connection agreements, scheduling protocols, and deviation settlement mechanisms must be urgently updated. AI campuses must be reclassified as "Large Special Consumers" with mandatory real-time NLDC telemetry and LVRT compliance for all power electronics >1 MW.

ENGINEERING MITIGATION STACK

BESS <100ms response | E-STATCOM ±200 MVAR | Demand Response to NLDC | GPU job scheduling API | Virtual Synchronous Machine | Active harmonic filters

GW Power Architecture, BESS / E-STATCOM & Green DC

National Grid **400–765 kV** → Campus GIS Substation **400 kV** → HV Ring **33/66 kV** → Data Hall Transformers **11 kV** → UPS / BESS **2N Topology** → GPU Rack **10–50 kW**

BESS — GRID STABILITY

DP 6

Response time	<100ms
Min. capacity	200 MWh
Discharge	30–60 min

- Primary frequency response — absorbs instantaneous load swing
- Smooths GPU training batch ramp-up/down
- LFP chemistry preferred; flow batteries for >4h buffer

E-STATCOM

DP 6

Reactive range	±200 MVAR
Response speed	<5ms

- Prevents voltage collapse on sudden loss of 400–500 MW campus load
- Integrated harmonic filtering — IEEE 519 THD compliance
- VSM mode: synthetic inertia emulation

GREEN DC TARGETS

DP 5

PUE ≤ 1.2	**CFE%** 90%+ hourly
WUE <0.5 L/kWh	**ZLD** Zero Liquid Discharge

500–800 MW captive solar + wind. SMR baseload by 2032+. 1 GW campus: ~20M L/day water — ZLD + liquid cooling (DLC/immersion) mandatory.

BRIDGING THE GENERATION & STORAGE GAP

DP 7

- **0–2 yr:** BESS + demand response + gas peakers as spinning reserve
- **2–5 yr:** Captive solar/wind + 4–8h BESS; must-run RE PPAs
- **5–10 yr:** SMR baseload + pumped hydro + green hydrogen for dispatchable capacity

CO-ORDINATED CONTROL

DP 8

Grid Signal NLDC/RLDC → Campus EMS/SCADA → BESS Controller + E-STATCOM VSC

GPU job scheduling co-ordinated with BESS SoC and generation dispatch — prevents compounding grid stress during concurrent AI training events across multiple campuses.

Solutions, Best Practices & Co-Investment Models

DP 1 & 2 — SITE SELECTION & CLUSTER STUDIES

Grid-First Site Selection + Mandatory Studies

Site selection must co-optimise: 400 kV proximity (<50 km), RE generation potential, watershed stress, seismic zone, submarine cable access. Full PSS/E + PSCAD cluster study mandatory for every campus >200 MW before CERC connection approval. 640+ acre contiguous parcels required for modular phased build-out.

DP 3 & 5 — RESOURCES & GREEN DC

Integrated Land–Energy–Water Planning

Paradigm shift from ad-hoc industrial zoning to dedicated AI Park SEZs with pre-approved grid, water allocations, and fast-track environmental clearances. Closed-loop DLC/immersion cooling + ZLD wastewater. CFE% tracking, PUE ≤ 1.2, WUE < 0.5 L/kWh as connection licence conditions.

DP 4 — REGULATION

Update Grid Code for GW-Class Consumers

CERC/SERC must create "Large AI Consumer" category: mandatory cluster studies, real-time telemetry to NLDC, LVRT compliance for all power electronics >1 MW, graduated deviation penalties to incentivise load predictability. DC-Transco joint grid investment agreements with long-term Use-of-System tariffs.

DP 8 — CO-INVESTMENT MODELS

Five Viable Business Structures

1. DC operator + TRANSCO joint venture for dedicated 400 kV line. **2.** Captive RE + grid export PPA (ISTS waiver). **3.** GPU scheduler enrolled in NLDC Ancillary Services Market (demand response revenue). **4.** AI Park SEZ with state-backed grid pre-investment. **5.** InvIT structure pooling transmission assets for institutional capital.

DP 6 & 7 — BESS / STATCOM & STORAGE GAP

Mandatory Stability Assets + 3-Horizon Plan

Min. BESS: 20% of IT load, 30-min duration, pre-qualified to CERC ancillary services spec. E-STATCOM: ±15% of peak load in MVAR. Horizon plan: gas peakers now → captive RE + 8h BESS (2–5 yr) → SMR + pumped hydro (5–10 yr). BESS + demand response converts grid's biggest threat into an ancillary services revenue stream.

⚠ UTILITY INVESTMENT DILEMMA — DP 8 & 10

The Trillion-Dollar Stranding Risk

Power utilities are re-evaluating grid investment plans given rapid on-site generation growth and emerging space-based DC prospects. If AI loads shift significantly off-grid, trillions in planned generation + grid strengthening risk becoming stranded assets. Regulators need demand forecasting frameworks that explicitly model AI workload elasticity and captive generation scenarios.

Future of Data Centres: Ground, Edge, or Orbit?

GROUND-BASED AI FACTORIES

- Proven technology; established supply chains
- GW-scale feasible but grid-constrained
- Water & land intensive — regulatory risk rising
- Best for: large-scale training + sovereign data requirements
- **India's primary play: 2025–2035**

SPACE-BASED DATA CENTRES

DP 10

- **Nov 2025:** NVIDIA H100 GPU deployed on SpaceX Starship — operational in LEO
- **Jan 2026:** Musk announces 1M satellites over 10 years
- Solar power 24×7 — no intermittency; cooling near-zero
- LEO latency 1–20ms — suitable for inference; training TBD
- Data uplink/downlink bandwidth remains a key constraint

GRID IMPLICATIONS OF SPACE DCS

- If training migrates to orbit → terrestrial campuses → inference-only → 60–70% load reduction
- Trillions in planned grid investment may become stranded
- Utilities need 3-scenario DC load forecasting: *base / captive-RE / space-shift*
- Grid connection agreements need off-ramp and re-use clauses
- Regulators must act on this uncertainty **now**

OPEN QUESTIONS FOR THE PANEL

- Which workloads are permanently terrestrial? (Sovereignty, real-time latency, bandwidth)
- Can space DCs handle full training runs, or inference only?
- What happens to India's $95B DC investments if orbit-scale is viable by 2032–2035?
- Who owns the "grid equivalent" (power infrastructure) in LEO?

Engineering Imperative: India must build ground-based AI Factories *and* simultaneously develop regulatory frameworks robust enough to adapt as space-based compute matures. Grid investment decisions made in 2025–2030 will define India's infrastructure posture for the next 30 years. The answer is not to wait — it is to design for optionality.

DP1 Site Selection · DP2 Cluster Studies · DP3 Resources · DP4 Regulations · DP5 Green DC · DP6 BESS/STATCOM · DP7 Storage Gap · DP8 Co-Investment · DP9 Best Practices · DP10 Space DCs